

MailTrace AI

Digital Email Forensic Investigation Report

1. Case Information

Case ID	MT-20260913181626-7220D2
Generated At	2026-09-13 18:16:33
Filename	bec_test.eml
Sender	Chief Executive Officer <ceo@executive-office.example>
Receiver	finance@example.com
Subject	Urgent Confidential Payment Request

2. Threat Risk Assessment

Risk Score	85.97 / 100
Risk Level	CRITICAL
Primary Threat	Bec
Recommended Action	Immediate containment, sender verification and forensic investigation recommended.

3. Machine Learning Analysis

Classification: Business Email Compromise

Confidence: 17.84%

Threat Class	Probability
Business Email Compromise	17.84%
Social Engineering	13.09%
Financial Fraud	10.39%
Legitimate Email	10.17%
Phishing	8.87%

4. Email Header Forensics

Authentication	Status
SPF	FAIL
DKIM	FAIL
DMARC	FAIL
Header Anomalies	5

5. Indicators of Compromise

IOC Type	Count
Emails	5
URLs	0
Domains	4

IP Addresses	0
MD5	0
SHA1	0
SHA256	0

6. Geo Intelligence

City: Bengaluru

Region: Karnataka

Country: India

Latitude: 12.9716

Longitude: 77.5946

Infrastructure: Demo Corporate Infrastructure

7. Explainable Findings

ML Threat Classification [HIGH] — ML model identified the email as Bec.

Header Forensics [HIGH] — Reply-To domain differs from sender domain.

Header Forensics [MEDIUM] — Return-Path domain differs from sender domain.

Header Forensics [HIGH] — SPF authentication failed.

Header Forensics [HIGH] — DKIM authentication failed.

Header Forensics [HIGH] — DMARC authentication failed.

Authentication [HIGH] — SPF authentication failed.

Authentication [HIGH] — DKIM authentication failed.

Authentication [HIGH] — DMARC authentication failed.

IOC Analysis [MEDIUM] — 9 indicators of compromise were extracted.

Content Analysis [MEDIUM] — Suspicious language patterns detected: Financial Fraud

Forensic Correlation [HIGH] — Multiple email authentication mechanisms failed.

Threat Correlation [HIGH] — Content analysis indicates a high-risk social-engineering or fraud-related pattern.

BEC Detection [CRITICAL] — Business Email Compromise indicators were detected.

Identity Forensics [HIGH] — Header anomalies correlate with failed sender authentication.

8. Threat Graph Summary

Graph Nodes: 8

Graph Relationships: 7

Forensic Notice: Geo intelligence represents approximate infrastructure-level or controlled demo location data. It must not be interpreted as an exact physical location of a sender.

Generated by MailTrace AI Threat Intelligence Engine